The Wayback Machine - https://web.archive.org/web/20241212211816/https://iowow.softmotions.com/wal

Write Ahead Log

Write ahead log (WAL) used to provide database data fault tolerance in the case of system/hardware failures. WAL recovery logs used in automatic recovery procedure get the most actual and consistent database state before a crash.

> NOTE: Activated WAL slows down db write operations by factor of 2

Savepoint

Flushing of *WAL* file to filesystem at consistent database state.
In the case of failure it can be rolled forward to restore database state to the latest savepoint time stamp.

Checkpoint

All pending *WAL* persisted database changes are applied to the database file.
WAL file data truncated to zero following by *savepoint* sync.
All durty pages flushed to disk.

Database WAL mode activated by the following `IWKV_OPTS.wal` settings:

```
IWKV_OPTS opts = {
    ...
    .wal = {
      .enabled = true,  // WAL Enabled/Disabled

      // Time period between WAL file sync at consistent database state. Defaut: 10 seconds
      .savepoint_timeout_sec = 10,

      // Time period between WAL checkpoints. Default: 5 min.
      .checkpoint_timeout_sec = 300,

      // Size of internal memory buffer for WAL operations. Default 8Mb
      .wal_buffer_sz = 8 * 1024 * 1024,

       // Maximum amount of memory used by duty pages. Overflow causes WAL checkpoint.
       // Default: 1Gb
      .checkpoint_buffer_sz = 1ULL * 1024 * 1024 * 1024
    }
  };
```

How WAL works

- Database file mmaped using private copy-on-write mapping `MAP_PRIVATE`

- All ongoing changes saved into WAL file then applied to privately mmaped database file region
- **Checkpoint performed on:**
  - Amount of private mmap changes data exceeds a specified limit: `.checkpoint_buffer_sz`
  - `.checkpoint_timeout_sec` timeout occuried
- **Checkpoint steps:**
  1. Perform *savepoint*
  2. Remap database file in shared mmap mode `MAP_SHARED`
  3. Apply all write WAL operations to the database file
  4. Restore copy-on-write mapping `MAP_PRIVATE` for database file.
  5. Truncate *WAL* to zero

Recovering

If database opened with WAL enabled option and found not empty WAL log on file system it will be applied to the database file until last successful *savepoint*.